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## RESEARCH ARTICLE

# Toward Generating Quality Test Questions and Answers Using Quantized Low-Rank Adapters in LLMs

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**ABSTRACT** Traditional approaches to question-and-answer generation are resource-intensive, necessitating innovative automation techniques to address these challenges. To this end, we propose fine-tuning strategies based on Quantized Low-Rank Adaptation (QLoRA), utilizing a meticulously curated domain-specific dataset. As a case study, we focused on the Korea College Scholastic Ability Test (KCSAT), introducing the KCSAT-ENG dataset, which comprises questions and answers from real and mock tests. Our approach involved fine-tuning the LLaMA-3-8B-Instruct model to generate questions and answers for 22 distinct task types. A key innovation of our work is the deployment of separate models for question-and-answer generation, leveraging a cross-verification process to enhance accuracy. To evaluate the QLoRA technique, we conducted extensive experiments by varying quantization, rank, and alpha values. The results highlighted optimal configurations: the question generation model performed best with  $rank, \alpha = 32, 8$  respectively without quantization, while the answer generation model achieved optimal results with  $rank, \alpha = 64, 16$  respectively. Compared to a non-fine-tuned LLaMA-3-8B-Instruct model, our question generation model demonstrated a 41.5% improvement, and the answer generation model achieved a 16.1% improvement. These findings underscore the potential of QLoRA-based fine-tuning in creating accurate, cost-effective, and scalable automated educational tools.

**INDEX TERMS** Natural language processing, large language model, text generation, parameter efficient fine tuning.

## I. INTRODUCTION

Large Language Models (LLM) have demonstrated remarkable performance in Natural Language Processing (NLP) tasks, excelling in various applications such as text generation, machine translation, and question answering [1]. However, these models often display suboptimal performance in domain-specific tasks when not explicitly fine-tuned or trained for such specialized purposes [2]. To address such issues, researchers have proposed several popular techniques, which include instruction fine-tuning [3], Retrieval-Augmented Generation (RAG) [4], and Low-Rank

Adaptation (LoRA) [5]. Instruction fine-tuning involves fine-tuning the model to better understand and follow instructions provided in natural language. RAG uses separate databases to complement knowledge to answer the given questions. LoRA directly adds more layers inside LLMs to capture domain-specific knowledge.

Over the past few decades, AI-assisted educational technology has garnered growing attention and become increasingly prevalent in diverse learning contexts [6], [7], [8], [9], [10], [11]. Meanwhile, LLMs have emerged as a transformative force in multiple domains, consistently achieving top-tier performance across educational tasks [12], [13], [14]. In particular, question generation has become a central research focus in applying LLMs to education, largely

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because it is widely used in instructional practice [15]. The task of Question Answering is one of the crucial use cases of LLMs. On the other hand, generating answers for education and training of students and apprentices in various fields is also important but has not been thoroughly explored yet. In this paper, we demonstrate how to improve LLMs to generate quality test questions and answers. We chose the Korea College Scholastic Ability Test (KCSAT), a national-level exam held annually by the Ministry of Education in South Korea, playing a crucial role in the college admission process. It is typically taken by high school seniors every November for university admission. In particular, the English section of the KCSAT is standardized in its task types and students are encouraged to practice many questions that follow the same task types as the exam. Generating a large number of high-quality questions for students requires significant cost and time.

In this paper, we adopt LLMs to automatically generate questions in the KCSAT-ENG format and provide answers, thereby delivering high-quality questions and answers to students while reducing the cost and time required for question and answer generation. Access to high-quality learning materials may be limited due to economic and geographical constraints. However, our study aims to provide more learners with equitable educational opportunities. Many researchers choose small language models for specific tasks. However, creating high-quality KCSAT-style questions and answers requires advanced reasoning and problem-solving abilities. A previous study [16] shows that small language models struggle with both complex reasoning tasks and problem-solving, whereas large language models consistently perform better. Since our approach involves maintaining a consistent format, creating well-structured questions, generating accurate answers, and providing detailed explanations, we have determined that large language models are the most suitable choice. We use instruction fine-tuning and Quantized Low-Rank Adaptation (QLoRA) [17] techniques while we exclude RAG because our goal is to generate entirely new questions and answers based solely on the model's capabilities.

We aim to generate high-quality test questions and answers using QLoRA in LLMs. Our contributions are as follows.

- We constructed a high-quality dataset of past English exam questions for the task of question and answer generation. This dataset strictly follows the unique format and answer structure of the KCSAT-ENG, enabling efficient data preprocessing and high-quality question generation through customized instructions and systematic prompt design. It includes definitions, examples, and Chain-of-Thought (CoT) reasoning for 22 distinct question types.
- We improved generation accuracy through a cross-validation process using two separate models for question and answer generation. In addition, we applied a post-validation regeneration step to enhance output quality, reinforcing both the accuracy and format conformity of the final results, thereby increasing the overall

reliability of the generated content. This design led to a significant boost in format conformity. Specifically, for question generation, the format verification pass rate reached 81.8%, representing a 41.5% improvement over the 57.8% achieved by the baseline LLaMA-3-8B-Instruct without instruction fine-tuning. For answer generation, the format verification rate was even higher, at 85.7%, showing a 17.4% improvement over the 73.0% baseline.

- We investigated the impact of QLoRA hyperparameters on the quality of generated questions and answers. A total of 19 QLoRA configurations were tested, varying rank, alpha, and quantization settings, yielding deep insights into optimal fine-tuning conditions. For instance, the highest format accuracy in question generation (81.8%) was achieved using non-quantized QLoRA with (rank=32,  $\alpha=8$ ). Similarly, the best BERTScore was obtained with the same configuration in answer generation. Furthermore, we developed a professional checklist to assess format compliance and explanation accuracy, aligning the evaluation process with educational goals beyond conventional NLP metrics, and applied it during model validation.

The paper is organized as follows: Section II describes related works. Section III explains the background of instruction fine-tuning and Parameter Efficient Fine Tuning (PEFT) methods used in the proposed KCSAT-ENG question and answer generation model. Section IV describes the dataset, which is newly proposed in this paper for generating KCSAT-ENG questions and answers, as well as the format of input data required for instruction fine-tuning and the structure of the question and answer generation model. Section V describes the composition of the test dataset, the experimental environment, and the evaluation metrics used for model performance evaluation experiments. Section VI is dedicated to the experimental evaluation and describes the achieved results.

## II. RELATED WORK

Prior studies in LLM have increasingly focused on leveraging LoRA and QLoRA to efficiently train LLMs specialized for specific domains. Such studies demonstrate that LoRA and QLoRA can achieve significant performance improvements with fewer parameters in domain-specific tasks such as translational, legal [18], financial [19], and medical [20], [21] contexts. Specifically in the education domain, research has been conducted on generating reading comprehension questions by fine-tuning GPT-2 [15]. However, no attempts have been made to apply parameter-efficient fine-tuning methods, such as LoRA or QLoRA, to this purpose. Our approach, fine-tuning LLaMA-3-8B models, is better suited for educational use because they can be tailored to specific domains like KCSAT-ENG, ensuring consistent question formats and answer styles. Unlike ChatGPT, they allow full parameter access, enabling fine-tuning for domain-specific knowledge and reasoning patterns. They can be deployed

locally, offering lower costs and better privacy for institutions. Additionally, they provide more reliable and reproducible outputs, which are critical for educational assessment and content generation.

This gap is particularly significant given that, while small language models (SLMs) are relatively easy to fine-tune, fine-tuning large language models (LLMs) with a large number of parameters poses significant challenges, such as high computational costs and resource limitations. To address these constraints and enable the efficient adaptation of LLMs for educational tasks, our study focuses on employing parameter-efficient fine-tuning methods, such as LoRA and QLoRA. Moreover, prior studies have not systematically tuned the hyperparameters of LoRA, specifically the rank and alpha, to identify their optimal values. Given the limitations of a single LLM in mastering multiple specialized domains [22], our study aims to develop an LLM specifically tailored to generate questions and answers for the English section of the KCSAT.

The task of question generation for LLMs has typically been achieved through prompt engineering [23]. While this approach allows for retaining model parameters, it may fail to achieve the high performance required in specialized domains, such as KCSAT question and answer generation. In this study, we collected a new dataset of KCSAT-ENG questions and answers and applied QLoRA-based fine-tuning. This method produced a domain-optimized model that offers comparable accuracy with significantly fewer parameters than ChatGPT.

CQuAE [24], one of the most similar works to ours, focused on developing a question-answer dataset tailored to the education domain and fine-tuning a model to generate inferential and analytical questions rather than simple fact-asking questions in French. Our study is distinguishable in terms of a wide range of question types, such as fill-in-the-blank and sentence ordering in the KCSAT-ENG format. Additionally, we designed a model that can restructure the input text into a question format and generate five choices for answer to align with the KCSAT-ENG question style. Besides the question-generation model, an answer-generation model was separately developed to provide detailed answers for each question, facilitating an intuitive understanding of the question-solving process.

Evaluation criteria for LLM outputs are critical in verifying the quality of the generated questions and answers. Prior studies, AQG systems [25] use automated metrics such as BLEU [26], ROUGE [27], and METEOR [28] useful for assessing grammatical accuracy and textual similarity of questions. Translation quality [29] and BERTScore [30] were also used to measure sentence similarity and accuracy. However, these automated metrics have limitations in sufficiently capturing contextual appropriateness or educational effectiveness. Biancini et al. [31] presents an evaluation framework for multiple-choice question generation using the criteria of compliance or sentence selection, clarity,

distractor selection, coherence of key selection, and learning utility. Similarly, Maity and Deroy [32] highlights the importance of human evaluation in assessing the alignment of generated questions with learning objectives, grammatical clarity, and content complexity and suggest the combined use of automated and human evaluation methods.

In this study, the answer to a question consists of two sections; translation and commentary. To overcome limitations in prior research, we systematically established evaluation criteria for assessing the quality of generated questions and answers. Due to challenges in evaluating educational usefulness and learning effectiveness, we focused on metrics that assess how accurately the generated outputs follow user instructions. We used automated evaluation metrics to assess the accuracy of the translation and ensure that the choice selected as the prediction in the generated commentary matches the ground truth, thereby ensuring the accuracy of the results.

### III. BACKGROUND

This section describes the background of instruction fine-tuning, LoRA, and related LLM techniques.

#### A. INSTRUCTION FINE-TUNING

In the task of generating English questions and answers, ensuring that the model accurately follows the user's instructions to produce questions in the correct format is essential. Since LLMs are trained to minimize contextual word prediction errors, there can be a misalignment between the model's training objective and the user's goal. To address this discrepancy, instruction fine-tuning has been proposed, which involves further training an LLM on pairs of instructions and corresponding outputs (Instruction, Output).

According to prior research [3], the primary benefits of instruction fine-tuning are as follows: (1) Fine-tuning an LLM on an instruction dataset helps bridge the gap between the next-word prediction objective of LLMs and the user's objective of instruction following. (2) It enables more controllable and predictable model behavior compared to standard LLMs, as instructions serve to constrain the model's outputs to align with desired response characteristics or domain knowledge, providing a mechanism for human intervention in model behavior. (3) Instruction fine-tuning is computationally efficient, allowing LLMs to quickly adapt to specific domains without extensive retraining or architectural modifications.

Leveraging these advantages, we designed custom instructions for generating questions and answers, paired with correct output examples as target outputs, and fine-tuned the model to enhance its adherence to user instructions. We meticulously created the KCSAT-ENG dataset, which consists of KCSAT-ENG-Question Generation (using actual KCSAT and mock test questions) and KCSAT-ENG-Answer Generation (utilizing detailed answers, generated by GPT-4o [33], for those questions). Since the number of answers

for the KCSAT-ENG and mock test questions was too small and simple, we used GPT-4o to create the answer dataset.

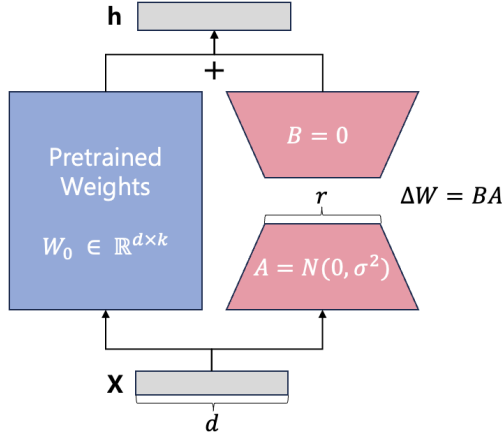


FIGURE 1. LoRA architecture.

### B. LORA

LoRA [5] is one of the Parameter-Efficient Fine-Tuning (PEFT) methodologies for LLM. LoRA reduces memory requirements by not updating all model parameters; instead, it adds a small number of trainable parameters (i.e., only adapters) [5]. LoRA freezes the weights of the pre-trained LLM and injects trainable Low-Rank decomposition matrices into the transformer architecture. This method significantly reduces the number of parameters that need to be trained for downstream tasks. For a parameter-efficient approach, the parameter increment  $\Delta\Phi = \Delta\Phi(\Theta)$  in (1) is encoded into a smaller parameter set  $\Theta$ . This parameter set efficiently represents  $\Delta\Phi$  and is much smaller in size compared to the pre-trained model's original parameter size  $\Phi_0$ . Therefore, the task of finding  $\Delta\Phi$  is transformed into an optimization problem over a smaller set of parameters  $\Theta$ , which can be represented by (1) [5].

$$\max_{\Theta} \sum_{(x,y) \in Z} \sum_{t=1}^{|y|} \log(p_{\Phi_0 + \Delta\Phi(\Theta)}(y_t | x, y_{<t})) \quad (1)$$

The structure of LoRA is shown in Fig. 1, where  $W_0 \in R^{d \times k}$  is the pre-trained weight matrix. The weight matrix update  $\Delta W$  is represented as the product of low-rank matrices  $B \in R^{d \times \text{rank}}$  and  $A \in R^{\text{rank} \times k}$ , where rank is used to reduce the dimensionality of the data represented by the matrix. During training,  $W_0$  is frozen, and only B and A are trainable parameters as described in (2) [5].

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x \quad (2)$$

Matrix A is initialized using a random Gaussian distribution, and B is initialized to zero. Afterward,  $\Delta W x$  is scaled by a scaling constant  $\frac{\alpha}{\text{rank}}$ , which means adjusting the strength when merging the low-rank matrix BA with the original weight  $W_0$ .

### C. QLoRA

Quantization [34] improves memory and computational efficiency by converting floating-point (FP32) values to low-bit integer values. However, during the quantization process, high-precision values like FP32 are converted to 8-bit or 4-bit integers, resulting in the loss of detailed information and subsequent dequantization can lead to further information loss, negatively affecting model performance.

To address these limitations effectively, QLoRA [17] introduces 4-bit NormalFloat (NF4), a quantization format optimized for normally distributed data, which minimizes information loss compared to conventional 4-bit integers and floats. Additionally, it incorporates Double Quantization, a method that reduces memory requirements by quantizing the quantization constants, leading to substantial storage savings. These innovations enable QLoRA to fine-tune large language models with minimal performance trade-offs, even on resource-constrained hardware.

In this study, we use QLoRA. QLoRA is an extended version of LoRA, which quantizes the pre-trained model to 4-bit, enabling even more memory-efficient fine-tuning. Despite the 4-bit quantization, QLoRA may conduct fine-tuning without performance degradation [17]. We perform model training in a low-memory setting by applying QLoRA.

### D. FLASH ATTENTION

Transformer models face the issue of quadratically increasing execution time and memory usage as sequence length increases. FlashAttention-2 [35] has been proposed to reduce memory complexity by improving attention computation-related mechanisms in three folds. First, The computation methods for the forward and backward pass of the attention mechanism are adjusted to reduce the number of non-matmul FLOPs. The second and third contributions are closely related to NVIDIA Graphic Processing Unit (GPU) programming. Second, attention computation is parallelized across different thread blocks in GPU, even for a single head, to increase occupancy. Third, work is distributed between warps within each thread block to reduce communication overhead through shared memory. In this study, FlashAttention-2 [35] is applied to optimize memory usage and attention computation time.

## IV. GENERATING QUALITY TEST QUESTIONS AND ANSWERS USING QLoRA

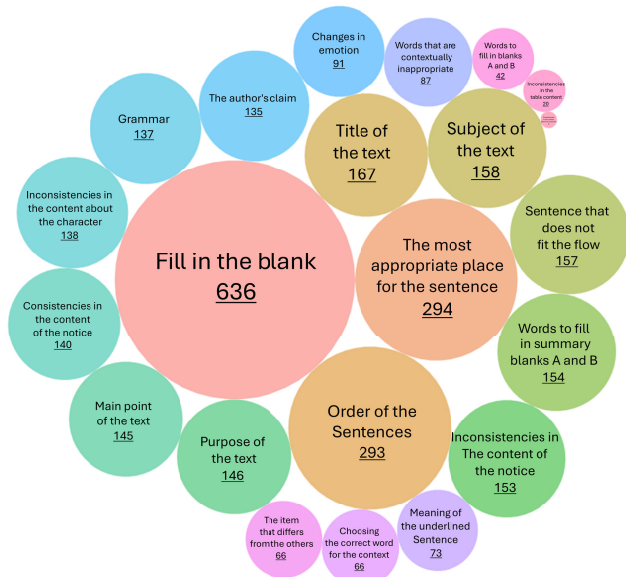
In this section, we describe our question-and-answer generation and our proposed model architectures and training methods.

### A. OUR TARGET TEST AND DATASET

Our target test is the English section of the Korea College Scholastic Ability Test. We call this KCSAT-ENG throughout this paper. We collected the KCSAT-ENG dataset ourselves from scratch. The English questions in the dataset were collected from 157 exam papers in KCSAT-ENG, which were administered by the Korea Institute for Curriculum &



Evaluation (KICE) and the Office of Education from 2014 to 2024. The total number of questions is 3,299 and the questions can be classified into 22 task types, as shown in Fig. 2.



**FIGURE 2.** Distribution of task types in collected CSAT dataset. Underlined number represents the number of tasks of each type.

The answer for a question is composed of (1) a translation of the question text into Korean and (2) a commentary of the ground truth. Since some exams did not provide Korean translations of the passages, we encountered difficulties in constructing the dataset. To address this issue, we generated the answer portion of the dataset using the GPT-4o API [36]. We provided prompts including collected questions, methods for commenting, and correct output examples. In other words, we use GPT-4o as a teller of ground truth for translation and commentary. We spent roughly five man-months collecting the dataset.

A prior study [37] showed that researchers achieved the highest performance when they composed instructions of definitions and two positive examples. In this paper, we use instructions composed of a definition and a single positive example. We developed the instruction for the “definition” in Question-and-Answer generation tasks, composed of (1) detailed instructions on how to create task-specific problems for question generation, and (2) clear guidelines on how to generate task-specific answers for answer generation.

However, the trained model did not always generate the intended output. To enhance the accuracy of the trained model, we also used the Chain-of-Thought (CoT) prompting strategy [38]. CoT helps LLM infer more accurate results through detailed intermediate reasoning steps. An example of CoT-based definitions in our prompts is shown in Fig. 3. On the other hand, our experiments showed that CoT-based definitions for answer generation negatively affected the

**Definition**

The provided text is for creating a type of question called “Sentence that does not fit the flow” for the English section of the CSAT. Follow the steps below to create a CSAT-style English question, text, and options:

**Step 1** - Use “Which sentence is irrelevant to the overall flow of the following text?” as the question.

**Step 2** - Develop one sentence that is unrelated to the flow of the text and randomly place it within the last five sentences of the provided text. Create options by listing one sentence from the end of the text in descending order.

**Step 3** - Provide the output in the following structure: “Question”: “Which sentence is irrelevant to the overall flow of the following text?” “Text”: [text]. “Options”: List of choices developed in Step 2. “(1) [Option1] (2) [Option2] (3) [Option3] (4) [Option4] (5) [Option5]” For example, if the following sentences are provided: In a highly commercialized setting such as the United States, it is not surprising that many landscapes are seen as commodities. In other words, they are valued because of their market potential. Residents develop an identity in part based on how the landscape can generate income for the community. This process involves more than the conversion of the natural elements into commodities. The landscape itself, including the people and their sense of self, takes on the form of a commodity. Over time, the landscape identity can evolve into a sort of “logo” that can be used to sell the stories of the landscape. Thus, California’s “Wine Country,” Florida’s “Sun Coast,” or South Dakota’s “Badlands” shape how both outsiders and residents perceive a place, and these labels build a set of expectations associated with the culture of those who live there. The output should be as follows: [“Question”: “Which sentence is irrelevant to the overall flow of the following text?”, “Text”: “In a highly commercialized setting such as the United States, it is not surprising that many landscapes are seen as commodities. In other words, they are valued because of their market potential. Residents develop an identity in part based on how the landscape can generate income for the community. (1) This process involves more than the conversion of the natural elements into commodities. (2) The landscape itself, including the people and their sense of self, takes on the form of a commodity. (3) Landscape protection in the US traditionally focuses on protecting areas of wilderness, typically in mountainous regions. (4) Over time, the landscape identity can evolve into a sort of “logo” that can be used to sell the stories of the landscape. (5) Thus, California’s “Wine Country,” Florida’s “Sun Coast,” or South Dakota’s “Badlands” shape how both outsiders and residents perceive a place, and these labels build a set of expectations associated with the culture of those who live there.”], “Options”: “(1) (2) (3) (4) (5)”, “ground\_truth”: “(3)”]

**FIGURE 3.** KCSAT-ENG’s prompt that describes how to create a question using the CoT technique for question generation. See Table 5 in Appendix A for the Korean original version of the prompt.

accuracy, so we decided not to employ the CoT strategy for answer generation as shown in Fig. 4.

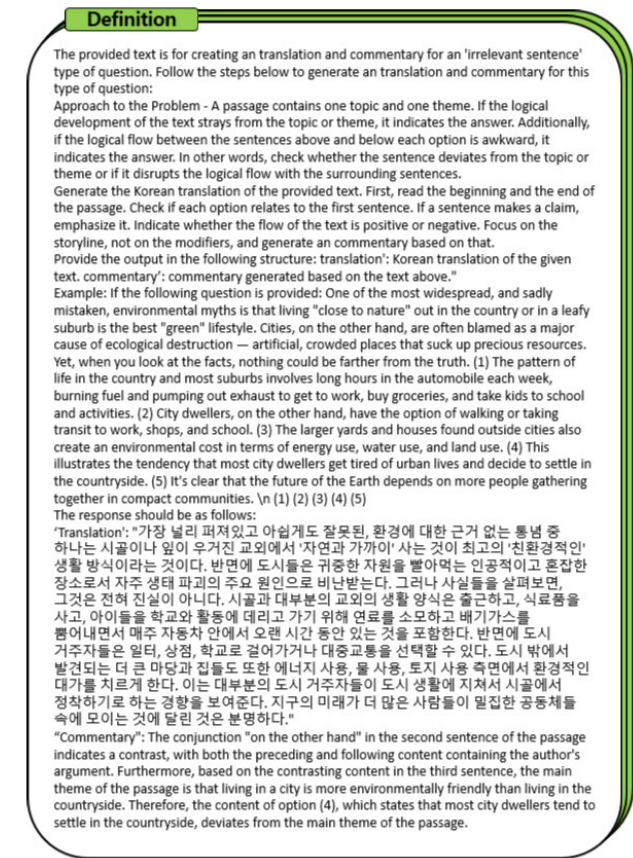
To prepare the KCSAT-ENG-Question Generation dataset as training data, each question is broken down into “question”, “original text”, “text”, “options” and “ground truth”, and it is stored in an “instance” field. The “original text” refers to the original text before being transformed into a question. For example, in the “Fill in the blank” task type, the original text represents the original text with the blanks filled in. On the other hand, regarding the KCSAT-ENG-Answer Generation dataset, each answer is broken down into “question”, “text”, “options”, and “answer”, and it is stored in an “instance” field. The “answer” includes translating the question into Korean and the commentary, which provides the ground truth and a detailed explanation of the question. The overall structure of KCSAT-ENG dataset is shown in Fig. 5.

## B. INSTRUCTION FINE-TUNING

We used instruction fine-tuning to guide the model in generating appropriate outputs for questions and answers in the KCSAT-ENG. The KCSAT-ENG-Question Generation dataset is transformed into three fields (definition, input, and responses) to train the question generation model. The details of the three fields are as follows.

- 1) Definition
- 2) Input: original\_text
- 3) Responses: question, text, options, ground truth

Similarly, the KCSAT-ENG-Answer Generation is transformed into three fields as follows.



**FIGURE 4.** KCSAT-ENG's prompt that describes how to solve a question for answer generation. See Table 5 in Appendix A for the Korean original version of the prompt.

- 1) Definition
- 2) Input: question, text, options
- 3) Responses: translation, commentary

The examples of training data is shown in Fig. 6.

### C. MODEL ARCHITECTURE

In this study, we employ LLaMA-3-8B-Instruct [39], [40], an open-source model, as our primary backbone. While GPT-4o offers excellent language understanding and generation capabilities, its closed nature prevents direct fine-tuning, restricting us to prompt-based few-shot techniques—an approach that is less suitable for long-term specialization in tasks such as KCSAT-ENG-style question generation. By contrast, LLaMA-3-8B-Instruct allows full access to model parameters in a local environment. Applying QLoRA enables partial fine-tuning, effectively internalizing domain-specific elements such as the structure of KCSAT-ENG reading questions and the placement of blank spaces. When the model incorporates such domain knowledge at the parameter level, it can generate questions and answers more consistently and reliably. These advantages led us to select LLaMA-3-8B-Instruct for question and answer generation. We composed the proposed system of two distinct models:

- 1) Question generation model: Responsible for generating high-quality questions based on the given input text.
- 2) Answer generation model: Provides translations and commentaries for the generated questions.

This division of tasks allows for greater specialization and optimization of each model for its respective purpose.

We may employ a validation process to verify the consistency between the prediction selected by the two models. The "prediction" refers to the answer generated by the model. Our system outputs the final result only when both models agree on an prediction. Suppose the probability of the question generation model selecting the ground truth is 0.6, and the probability of the answer generation model selecting the ground truth is 0.8. In this case, the probability that the final output is the ground truth increases to approximately 0.96. In this way, our system can achieve significantly higher accuracy compared to relying on a single model. Our system leverages the strengths of LLaMA-3-8B-Instruct, which excels in tasks that require adherence to human instructions and effective interpretation of sentence intent. Additionally, we employ QLoRA and FlashAttention-2 to address the challenges of computational time and memory usage in attention layers. More specifically, our study adopts the same strategy as prior research [5], [17] for implementing our model, ensuring efficient fine-tuning and optimized performance. The prior research demonstrated strong performance by applying LoRA to the attention projection matrices  $W_q$ ,  $W_v$  among  $W_q$ ,  $W_k$ ,  $W_v$ ,  $W_o$ . Furthermore, our study incorporates 4-bit QLoRA with the NF4 data type, building on prior research to achieve performance comparable to 16-bit full fine-tuning and 16-bit LoRA fine-tuning. Accordingly, our model architecture is illustrated in Figure 7.

## V. EXPERIMENTAL METHODOLOGY

In this section, we explain how we utilized the dataset and implemented our LLMs. We also outline our evaluation method.

### A. TEST DATASET

We conducted a performance test on the question generation model using a new dataset comprising 14 original texts from the 2023 KCSAT-related Korean textbook "KCSAT completion" [41], which were not included in the training data, as shown in Table 1. Similarly, we conducted a performance test on the answer generation model using a new dataset comprising 21 questions from the 2023 KCSAT-related textbook "KCSAT completion" and the Korea Military Academy entrance exam [42], as shown in Table 2. Both sources follow a format similar to the KCSAT-ENG and were also not included in the training data.

### B. IMPLEMENTATION DETAILS

The experiments in this study were conducted on a single server equipped with an AMD Ryzen Threadripper PRO 3955WX CPU (16 cores), 251 GB of system RAM, and

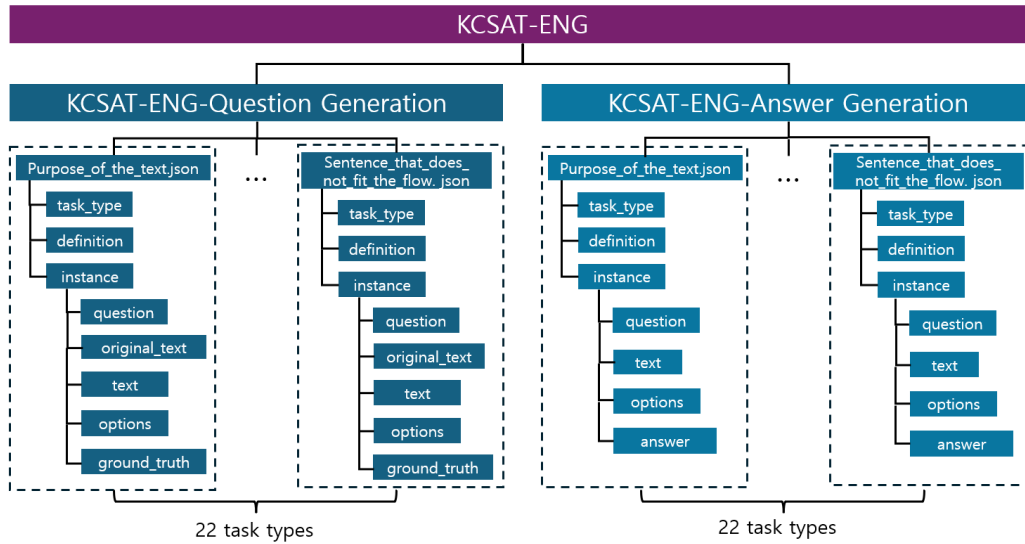


FIGURE 5. Structure of KCSAT-ENG dataset.

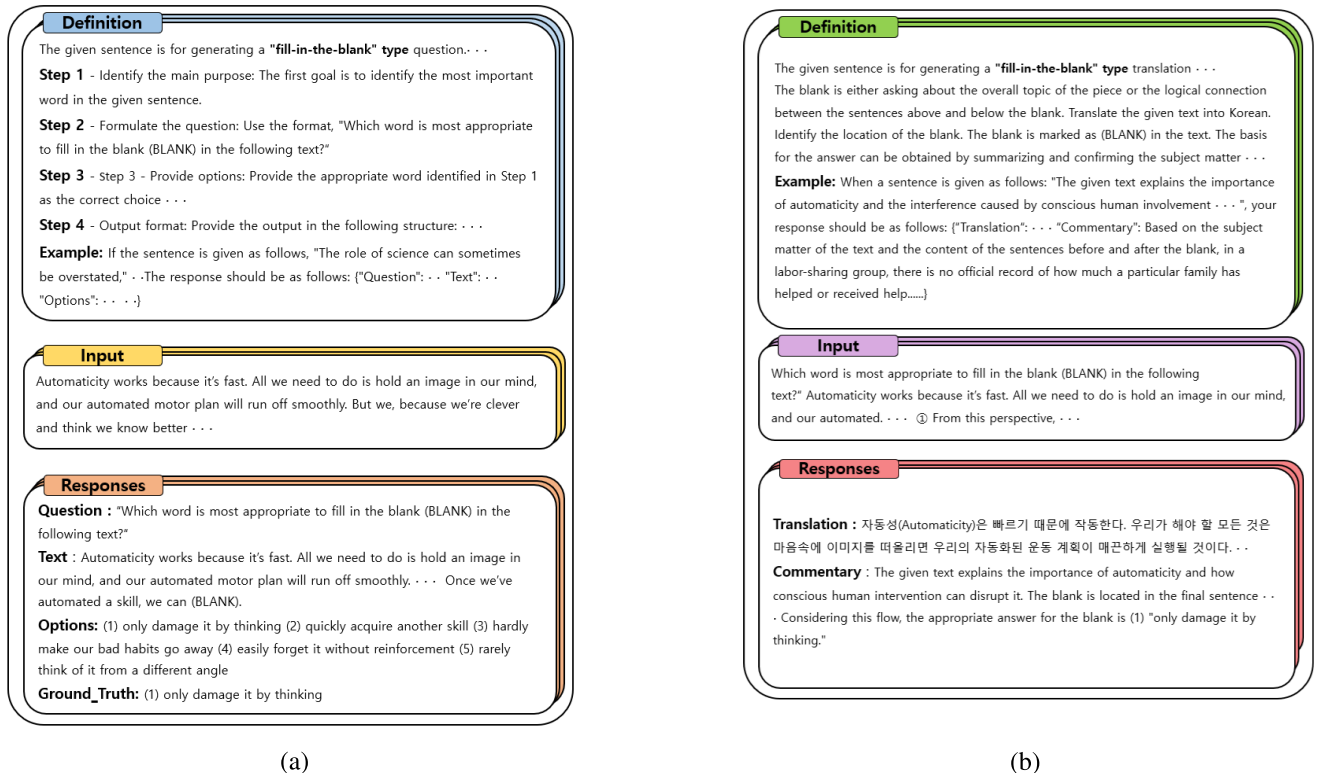


FIGURE 6. (a) Input example for model training for question generation. (b) Input example for model training for answer generation. See Table 5 in Appendix A for the Korean original of the input format.

three GPUs: one NVIDIA RTX 6000 Ada Generation GPU with 48 GB of VRAM and two NVIDIA GeForce RTX 3090 GPUs with 24 GB of VRAM each. For the fine-tuning process, we utilized the RTX 6000 Ada GPU, where the maximum VRAM usage was approximately 40 GB. The server was based on the Pro WS WRX80E-SAGE SE WIFI

motherboard. All implementations were performed using Python, and model training and evaluation were carried out using the PyTorch framework.

We applied QLoRA [17], which quantizes weights into 4-bit precision, only to the Query and Value projection matrices. In a previous LoRA study [5], the GPT-3 175B model

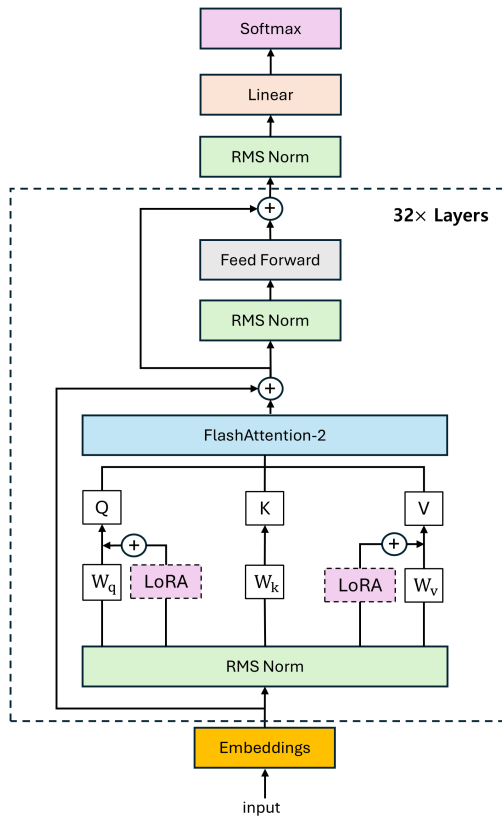


FIGURE 7. Model architecture.

TABLE 1. Elements of each task type that make up the test dataset for the question generation model.

| Representative type                   | Applicable type                                                                                                     |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Understanding the main idea           | Purpose of the text,<br>The author's claim,<br>Main point of the text,<br>Subject of the text,<br>Title of the text |
| Change in emotions                    | Changes in emotion                                                                                                  |
| Meaning of the underlined sentence    | Meaning of the underlined sentence                                                                                  |
| Table                                 | Inconsistencies in the table content                                                                                |
| Content about the character           | Consistencies in the content about the character,<br>Inconsistencies in the content about the character             |
| Notice                                | Consistencies in the content of the notice,<br>Inconsistencies in the content of the notice                         |
| Grammar                               | Grammar                                                                                                             |
| Vocabulary                            | Words that are contextually inappropriate,<br>Choosing the correct word for the context                             |
| Blank                                 | Fill in the blank,<br>Words to fill in blanks A and B                                                               |
| Sentence that does not fit the flow   | Sentence that does not fit the flow                                                                                 |
| Order of the Sentences                | Order of the sentences                                                                                              |
| Inserting sentences                   | The most appropriate place for the sentence                                                                         |
| Words to fill in summary              | Words to fill in summary blanks A and B                                                                             |
| The item that differs from the others | The item that differs from the others                                                                               |

demonstrated strong performance at ranks of 4 and 8 when the scaling factor ( $\alpha/\text{rank}$ ) was set to 1. Drawing on the findings of the LoRA and QLoRA studies, where scaling factors were set to 0.25, 1, and 2, we conducted extensive experiments adjusting  $\alpha$  to achieve scaling factors of 0.25, 1, and 2 for each rank. The combinations of hyperparameters, including rank,  $\alpha$ , and quantization, along with the

TABLE 2. Elements of each task type that make up the test dataset for the answer generation model.

| Task type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose of the text,<br>The author's claim,<br>Main point of the text,<br>Subject of the text,<br>Title of the text,<br>Changes in emotion,<br>Meaning of the underlined sentence,<br>Inconsistencies in the table content,<br>Inconsistencies in the content about the character,<br>Consistencies in the content of the notice,<br>Inconsistencies in the content of the notice,<br>Grammar,<br>Words that are contextually inappropriate,<br>Choosing the correct word for the context,<br>Fill in the blank,<br>Sentence that does not fit the flow,<br>Order of the sentences,<br>The most appropriate place for the sentence,<br>Words to fill in summary blanks A and B, |
| Words to fill in blanks A and B,<br>The item that differs from the others                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

corresponding percentage of trainable parameters, are shown in Table 3.

We used the Paged AdamW Optimizer [17] when we applied quantization; otherwise, we used the AdamW torch optimizer [43] for fine-tuning. We set the batch size to the maximum possible value that avoided out-of-memory (OOM) errors during training — 1 without quantization and 5 with quantization. Additionally, we set the number of epochs to 6, the learning rate to  $2e-4$ , the scheduler to constant, the warm-up ratio to 0.03, and the maximum sequence length to 2048.

### C. EVALUATION METHOD

In this study, we performed the format verification and assessed the accuracy rate through human evaluation, involving the authors of this paper.

All processes are objective, not subjective. Format verification tests whether the model generates questions and answers in the correct format according to the definition, represented as a binary outcome based on Equation (3), and analyzed as a percentage of true ( $=1$ ) occurrences. For example, in the case of the “Fill in the blank” question type, if the generated question satisfies the evaluation criteria such as “Does it classify and output the question, text, options, and prediction separately?” and “Are there five non-duplicate options (excluding the one that differs from the others)?”, but fails to meet the criterion “Is the blank in the text represented by ‘(BLANK)’?”, it is categorized as “any criteria not met.” As a result, the evaluator assigns a score of 0 to the generated question. This indicates that the question was generated incorrectly, and the model’s performance is evaluated based on the proportion of correctly generated questions out of the total number of generated questions. The evaluation criteria



**TABLE 3.** Combination of rank, alpha, and quantization for each test case with the corresponding ratio of trainable parameters.

| Test Case | Trainable Param(%) | Rank | Alpha | Scaling Factor | Quantization |
|-----------|--------------------|------|-------|----------------|--------------|
| 1         | 0.01               | 2    | 2     | 1              | X            |
| 2         | 0.04               | 8    | 8     | 1              | X            |
| 3         | 0.08               | 16   | 4     | 0.25           | X            |
| 4         | 0.08               | 16   | 16    | 1              | X            |
| 5         | 0.08               | 16   | 32    | 2              | X            |
| 6         | 0.17               | 32   | 8     | 0.25           | X            |
| 7         | 0.17               | 32   | 32    | 1              | X            |
| 8         | 0.17               | 32   | 64    | 2              | X            |
| 9         | 0.34               | 64   | 16    | 0.25           | X            |
| 10        | 0.34               | 64   | 64    | 1              | X            |
| 11        | 0.34               | 64   | 128   | 2              | X            |
| 12        | 0.02               | 2    | 2     | 1              | O            |
| 13        | 0.02               | 2    | 4     | 2              | O            |
| 14        | 0.04               | 4    | 1     | 0.25           | O            |
| 15        | 0.04               | 4    | 4     | 1              | O            |
| 16        | 0.04               | 4    | 8     | 2              | O            |
| 17        | 0.07               | 8    | 2     | 0.25           | O            |
| 18        | 0.07               | 8    | 8     | 1              | O            |
| 19        | 0.07               | 8    | 16    | 2              | O            |

for each task type used in the experimental evaluation are described in Table 6 and Table 7 in Appendix B.

For accuracy rate, we compared the generated prediction of the answer model to the ground truth for accuracy. We used the F1 score of BERTScore ( $F_{BERT}$ ) [30] to evaluate whether the model provides accurate translations.

$$y = \begin{cases} 1, & \text{all criteria met} \\ 0, & \text{any criteria not met} \end{cases} \quad (3)$$

## VI. EXPERIMENTAL RESULTS

In this section, we describe the experimental results for format verification,  $F_{BERT}$ , and accuracy rate based on the hyperparameter combinations, including rank, alpha, and quantization, as shown in Table 4. We compare these results with LLaMA-3-8B-Instruct without instruction fine-tuning and GPT-4o. For the evaluation results by task type, see Fig. 9 and Fig. 10 in Appendix C.

### A. QUESTION GENERATION

We evaluate the format verification of the generated questions through human evaluation. For question generation, the average format verification of the non-quantized group is 78.0%, while that of the quantized group is 68.1%, resulting in a 14.5% improvement when not quantized. The experimental results demonstrated that format verification according to rank and alpha does not exhibit a specific pattern, as shown in Fig. 8. To summarize, when we apply non-quantization with rank and alpha set to 32 and 8, respectively, the format verification pass rate reaches 81.8%, representing a 41.5%

improvement compared to 57.8% achieved by LLaMA-3-8B-Instruct without instruction fine-tuning, and a 2.8% improvement compared to 79.6% achieved by GPT-4o. If the generated output lacks the basic structure of a question, it can be regenerated through a filtering process to enhance format verification. In 247 test cases, the output was generated with an average of 1.08 regenerations.

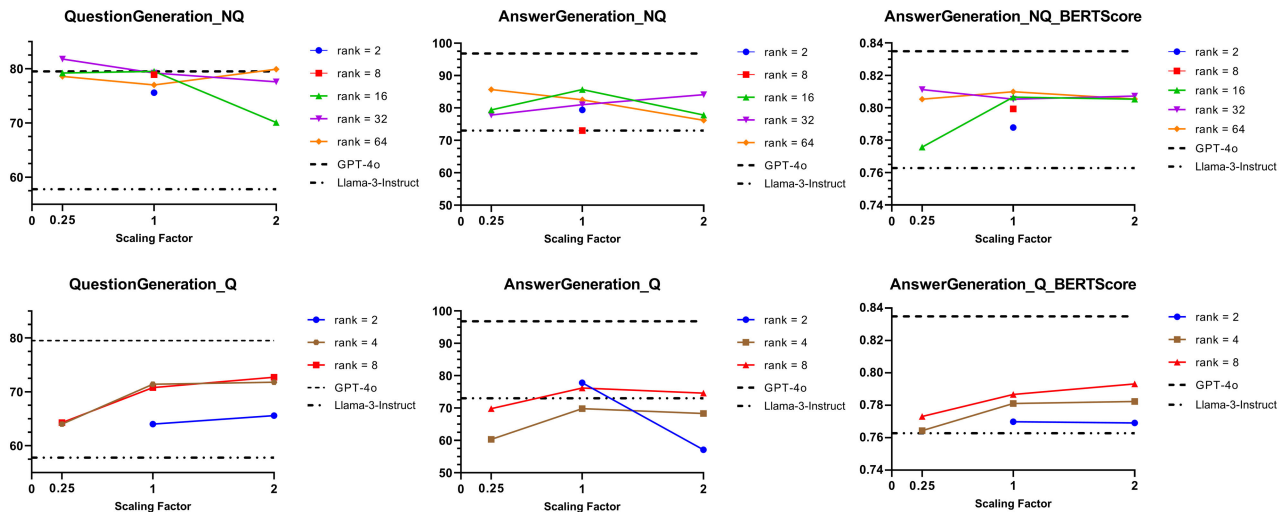
In addition to these findings, we compared the computational efficiency in terms of forward FLOPs per token across different model architectures. GPT-3, with its 175 billion parameters [44], requires 2 FLOPs per parameter per token for forward processing, resulting in a computational load of  $175B \times 2 \times k$  FLOPs for  $k$  tokens. For LLaMA-3, which utilizes 8 billion parameters, the forward FLOPs are calculated as  $8B \times 2 \times k$ . Notably, GPT-4o is presumed to utilize an even larger parameter set than GPT-3. Our model, with significantly fewer parameters than GPT-3 (over 20 times fewer), achieves superior performance with substantially reduced forward FLOPs. This efficiency demonstrates that our model not only matches but also outperforms larger models such as GPT-4o in format verification, underscoring the advantages of our parameter-efficient architecture.

### B. ANSWER GENERATION

We evaluate the format verification of the generated answers through human evaluation, assess the translation quality of the answers using BERTScore [30], and verify the accuracy of the answers generated by the answer model through alignment with the ground truth.

**TABLE 4.** Performance metrics across various hyperparameter configurations compared to non-fine-tuned LLaMA-3-8B-Instruct and GPT-4o. For each test case, the first column lists the format verification, the second shows the  $F_{BERT}$  for answer generation, and the third provides the accuracy rate. Bold indicates the highest performance metrics, while underlined indicates the second-best.

| Test Case<br>(rank, alpha) | Quantization | Problem Generation | Explanation Generation |               |             |
|----------------------------|--------------|--------------------|------------------------|---------------|-------------|
|                            |              | format             | format                 | $F_{BERT}$    | $C_{rate}$  |
| 1 (2, 2)                   | x            | 75.7               | 79.4                   | 0.7878        | 58.7        |
| 2 (8, 8)                   |              | 78.9               | 73.0                   | 0.7993        | 54.0        |
| 3 (16, 4)                  |              | 79.2               | 79.4                   | 0.7757        | 65.0        |
| 4 (16, 16)                 |              | 79.6               | <b>85.7</b>            | 0.8065        | 60.3        |
| 5 (16, 32)                 |              | 70.1               | 77.8                   | 0.8053        | <b>73.0</b> |
| 6 (32, 8)                  |              | <b>81.8</b>        | 77.8                   | <b>0.8112</b> | 69.8        |
| 7 (32, 32)                 |              | 79.2               | 81.0                   | 0.8053        | <b>73.0</b> |
| 8 (32, 64)                 |              | 77.6               | <u>84.1</u>            | 0.8072        | 68.3        |
| 9 (64, 16)                 |              | 78.6               | <b>85.7</b>            | 0.8053        | <u>71.4</u> |
| 10 (64, 64)                |              | 77.0               | 82.5                   | <u>0.8099</u> | <u>71.4</u> |
| 11 (64, 128)               |              | <u>79.9</u>        | 76.2                   | 0.8054        | 61.9        |
| 12 (2, 2)                  | o            | 64.0               | 77.8                   | 0.7699        | 60.3        |
| 13 (2, 4)                  |              | 65.6               | 57.1                   | 0.7691        | 34.9        |
| 14 (4, 1)                  |              | 64.0               | 60.3                   | 0.7644        | 44.4        |
| 15 (4, 4)                  |              | 71.4               | 69.8                   | 0.7811        | 46.0        |
| 16 (4, 8)                  |              | 71.8               | 68.3                   | 0.7824        | 50.8        |
| 17 (8, 2)                  |              | 64.3               | 69.8                   | 0.7731        | 47.6        |
| 18 (8, 8)                  |              | 70.8               | 76.2                   | 0.7867        | 55.6        |
| 19 (8, 16)                 |              | 72.7               | 74.6                   | 0.7932        | 57.1        |
| LLaMA-3-Instruct           | -            | 57.8               | 73.0                   | 0.7628        | 46.0        |
| GPT-4o                     |              | 79.6               | 96.8                   | 0.8349        | 88.9        |



**FIGURE 8.** Performance comparison: Our models vs non-fine-tuned LLaMA-3-8B-Instruct vs GPT-4o.

For answer generation, the average format verification of the non-quantized group is 80.2%, whereas that of the quantized group is 69.2%, resulting in a 15.9% improvement when not quantized. If the generated output lacks the basic structure of an answer, it can be repeatedly regenerated through a filtering process to enhance format verification. In 247 test cases, the output was generated after an average

of 1.45 regenerations. Among these, 16 cases exceeded the maximum regeneration limit, which was set to 7.  $F_{BERT}$  was 0.8017 for the non-quantized group and 0.7775 for the quantized group, indicating a 3.1% improvement with quantization. Accuracy was 66.09% for the non-quantized group and 49.60% for the quantized group, showing a 33.3% improvement when not quantized. The experimental results

**TABLE 5.** Example of question generation and answer generation definition corresponding to the Fill in the blank type.

| Purpose             | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question Generation | <p>주어진 글은 '빈칸에 들어갈 말' 유형 질문을 생성하기 위한 것입니다. 주어진 글을 사용하여 '빈칸에 들어갈 말' 유형의 수능 영어 스타일 질문, 본문, 보기, 답을 다음 단계에 따라 생성하세요. 단계 1 - 주어진 글에서 가장 중요한 단어 또는 구문을 파악하여 (BLANK)로 대체하세요. 단계 2 - '다음 글의 빈칸(BLANK)에 들어갈 말로 가장 적절한 것은?'을 질문으로 사용하세요. 단계 3 - 단계 1에서 찾은 알맞은 단어 또는 구문을 정확한 선택지로 제시하고, 문맥과 일치하지 않는 네 가지 부정확한 선택지를 제시하세요. 단계 4 - 다음 구조로 출력을 제공하세요. "질문": "다음 글의 빈칸 (BLANK)에 들어갈 말로 가장 적절한 것은?". "본문": 단계1의 지침을 따르는 본문. "보기": 단계3에서 제시된 선택지 목록, "(1) {보기1} (2) {보기2} (3) {보기3} (4) {보기4} (5) {보기5}". "답": 단계3에서 선택된 올바른 선택지. 예: 다음과 같이 문장이 주어진 경우, The role of science can sometimes be overstated, with its advocates slipping into scientism. Scientism is the view that the scientific description of reality is the only truth there is. With the advance of science, there has been a tendency to slip into scientism, and assume that any factual claim can be authenticated if and only if the term 'scientific' can correctly be ascribed to it. The consequence is that non-scientific approaches to reality—and that can include all the arts, religion, and personal, emotional and value-laden ways of encountering the world—may become labelled as merely subjective, and therefore of little account in terms of describing the way the world is. The philosophy of science seeks to avoid crude scientism and get a balanced view on what the scientific method can and cannot achieve. 반응은 다음과 같아야 합니다: {"질문": "다음 글의 빈칸(BLANK)에 들어갈 말로 가장 적절한 것은?", "본문": "The role of science can sometimes be overstated, with its advocates slipping into scientism. Scientism is the view that the scientific description of reality is the only truth there is. With the advance of science, there has been a tendency to slip into scientism, and assume that any factual claim can be authenticated if and only if the term 'scientific' can correctly be ascribed to it. The consequence is that non-scientific approaches to reality—and that can include all the arts, religion, and personal, emotional and value-laden ways of encountering the world—may become labelled as merely subjective, and therefore of little (BLANK) in terms of describing the way the world is. The philosophy of science seeks to avoid crude scientism and get a balanced view on what the scientific method can and cannot achieve.", "보기": "(1) question (2) account (3) controversy (4) variation (5) bias", "답": "(2)"}</p>                                         |
| Answer Generation   | <p>주어진 글은 '빈칸' 유형의 문제에 대한 해석 및 해설을 작성하기 위한 것이다. 주어진 글을 사용하여 '빈칸' 유형의 문제에 대한 해석 및 해설을 아래 설명에 따라 생성하세요. 문제 접근법 - 글 하나에는 주제가 하나입니다. 빈칸은 글 전체의 주제, 유지를 묻는 것 아니면, 논리적 전개상 빈칸 위, 아래 문장과의 관계를 묻는 문제입니다. 주어진 글의 한국어 해석을 생성하세요. 빈칸 위치를 파악하세요. 빈칸은 글에서 (BLANK)로 표시됩니다. 빈칸 위치가 첫 문장일 경우, 출제 의도는 주제입니다. 답의 근거는 결론과 주장문에서 찾을 수 있습니다. 빈칸 위치가 마지막 문장일 경우, 출제 의도는 결론을 찾는 것입니다. 답의 근거는 소재와 주장문, 빈칸의 앞 문장에서 찾을 수 있습니다. 빈칸 위치가 가운데인 경우에, 출제 의도는 불분명합니다. 답의 근거는 소재, 주장문, 빈칸 위, 아래 문장, 결론을 모두 요약 확인하여 얻을 수 있습니다. 동사와 연결어를 파악하고, 예시 앞, 자문-자답, 실험 연구 결과 등을 파악하여 주장문을 찾으세요. 다음 구조로 출력을 제공하세요. "해석": 주어진 글의 한국어 해석. "해설": 위 글을 바탕으로 생성한 해설. 예: 다음과 같은 문장이 주어진 경우: In labor-sharing groups, people contribute labor to other people on a regular basis (for seasonal agricultural work such as harvesting) or on an irregular basis (in the event of a crisis such as the need to rebuild a barn damaged by fire). Labor sharing groups are part of what has been called a "moral economy" since no one keeps formal records on how much any family puts in or takes out. Instead, accounting is (BLANK). The group has a sense of moral community based on years of trust and sharing. In a certain community of North America, labor sharing is a major economic factor of social cohesion. When a family needs a new barn or faces repair work that requires group labor, a barn-raising party is called. Many families show up to help. Adult men provide manual labor, and adult women provide food for the event. Later, when another family needs help, they call on the same people. (1) legally established (2) regularly reported (3) socially regulated (4) manually calculated (5) carefully documented 반응은 다음과 같아야 합니다: {"해석": "노동력 공유 집단에서 사람들은 정기적으로(수확과 같은 계절적인 농사일을 위해) 혹은 비정기적으로(화재로 손상된 헛간을 다시 지어야 하는 것과 같은 위기 상황 발생 시) 다른 사람들에게 노동력을 제공한다. 아무도 어떤 가족이 얼마나 많이 투입하고 얼마나 많이 가져갔는지에 대해 공식적인 기록을 남기지 않으므로, 노동력 공유 집단은 '도덕적 경제'라고 불려 온 것의 일부다. 대신에, 정산은 사회적으로 규제된다. 그 집단은 다년간의 신뢰와 나눔을 바탕으로 하는 도덕적 공동체 의식을 가지고 있다. 북미의 특정 지역 사회에서는 노동력 공유가 사회적 응집성의 주요 경제적 요소이다. 한 가족이 새 헛간이 필요하거나 단체 노동력을 요하는 수리 작업에 직면할 때, 헛간 조성 모임이 소집된다. 여러 가족이 도우러 온다. 성인 남성은 육체노동을 제공하고, 성인 여성은 행사를 위한 음식을 제공한다. 나중에, 다른 가족이 도움이 필요할 때 그들은 같은 사람들을 부른다.", "해설": "위 글의 소재와 빈칸 앞 뒤 문장 등의 내용으로 미루어 보아 노동력 공유 집단에서는 어느 가족이 얼마나 도와주고 도움을 받았는지에 대한 공식적인 기록을 하지 않고, 신뢰와 나눔을 바탕으로 한 공동체에서 도움이 필요할 때마다 모여서 서로 도와주는 것이므로, 빈칸에 들어갈 말로 가장 적절한 것은 (3) '사회적으로 규제된다'이다. 정답을 찾고 이해하는 것 보다 오답을 지우는 것이 핵심이다.</p> |

demonstrated that format verification and  $F_{BERT}$  according to rank and alpha do not exhibit a specific pattern, as shown in Fig. 8.

In conclusion, without quantization and with rank,  $\alpha = \{16, 16\}, \{64, 16\}$ , format verification achieved 85.7%, which is a 17.4% improvement compared to the

73.0% of LLaMA-3-8B-Instruct without instruction fine-tuning, but 11.5% lower than the 96.8% of GPT-4o.  $F_{BERT}$  was highest at 0.8112 when rank,  $\alpha = 32, 8$  without quantization, showing a 6.4% improvement compared to the 0.7628 of LLaMA-3-8B-Instruct without instruction fine-tuning, but 2.8% lower than the 0.8349 of GPT-4o.

**TABLE 6. Evaluation Criteria about question generation. The numbers in parentheses indicate the number of task types.**

| Common criteria                                                                                                                              |                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does it classify and output the question, text, options and prediction separately?                                                           |                                                                                                                                                                                                                                                                                                                                    |
| Are there 5 non-duplicate options? (The item that is different from the others is excluded)                                                  |                                                                                                                                                                                                                                                                                                                                    |
| Task Type (22)                                                                                                                               | Additional criteria                                                                                                                                                                                                                                                                                                                |
| Purpose of the text (P1),<br>The author's claim (P2),<br>Main point of the text (P3),<br>Subject of the text (P4),<br>Title of the text (P5) | <ul style="list-style-type: none"> <li>No additional criteria</li> </ul>                                                                                                                                                                                                                                                           |
| Change in emotions (P6)                                                                                                                      | <ul style="list-style-type: none"> <li>Are all the options formatted as <i>emotion</i>-&gt;<i>emotion</i>?</li> </ul>                                                                                                                                                                                                              |
| Meaning of the underlined sentence (P7)                                                                                                      | <ul style="list-style-type: none"> <li>Is the sentence in the text wrapped with the underline token &lt;U&gt;&lt;/U&gt;?</li> </ul>                                                                                                                                                                                                |
| Inconsistencies in the table content (P8)                                                                                                    | <ul style="list-style-type: none"> <li>Does the text contain table format?</li> </ul>                                                                                                                                                                                                                                              |
| Consistencies in the content about the character (P9),<br>Inconsistencies in the content about the character (P10)                           | <ul style="list-style-type: none"> <li>Depending on the type of question, is the question 'What is consistent about ~?' or 'What is inconsistent about ~?'?</li> </ul>                                                                                                                                                             |
| Consistencies in the content of the notice (P11),<br>Inconsistencies in the content of the notice (P12)                                      | <ul style="list-style-type: none"> <li>Is the question '~What matches the contents of the guidance?' or '~What does not match the contents of the guidance?'?</li> </ul>                                                                                                                                                           |
| Grammar (P13)                                                                                                                                | <ul style="list-style-type: none"> <li>Are the words labeled (1) to (5) in the text wrapped with &lt;&gt;?</li> <li>Do the words corresponding to each number in the text and options match?</li> </ul>                                                                                                                            |
| Words that are contextually inappropriate (P14)                                                                                              | <ul style="list-style-type: none"> <li>Are numbers (1) to (5) labeled with underlined tokens in the text?</li> </ul>                                                                                                                                                                                                               |
| Choosing the correct word for the context (P15)                                                                                              | <ul style="list-style-type: none"> <li>Do (A), (B), and (C) exist in the text?</li> <li>Are (A), (B), and (C) in the text expressed in the form of word/word?</li> <li>Are the options expressed in the form of word...word...word?</li> <li>Are the words shown in the options also present in text (A), (B), and (C)?</li> </ul> |
| Fill in the blank (P16)                                                                                                                      | <ul style="list-style-type: none"> <li>Is the blank in the text represented by "(BLANK)"?</li> </ul>                                                                                                                                                                                                                               |
| Words to fill in blanks A and B (P17)                                                                                                        | <ul style="list-style-type: none"> <li>Do both (A) and (B) exist in the text?</li> <li>Are the options formatted as (A)word...(B)word?</li> </ul>                                                                                                                                                                                  |
| Sentence that does not fit the flow (P18)                                                                                                    | <ul style="list-style-type: none"> <li>Are numbers (1) to (5) labeled in the text?</li> <li>Are the options in the format of (1) (2) (3) (4) (5) (The options must not contain any text)?</li> </ul>                                                                                                                               |
| Order of the sentences (P19)                                                                                                                 | <ul style="list-style-type: none"> <li>Are A, B, and C properly placed in the text?</li> <li>Do all of A, B, and C exist in the options?</li> </ul>                                                                                                                                                                                |
| The most appropriate place for the sentence (P20)                                                                                            | <ul style="list-style-type: none"> <li>Are positions ((1))~((5)) marked in the text to insert sentences?</li> <li>Are the given sentences separated by "\n"?</li> <li>Are the options in the format of (1) (2) (3) (4) (5) (The options must not contain any text)?</li> </ul>                                                     |
| Words to fill in summary blanks A and B (P21)                                                                                                | <ul style="list-style-type: none"> <li>Is there a summary enclosed within '&lt; &gt;' at the end of the text?</li> <li>Is the summary a newly generated sentence?</li> <li>Does the summary contain both (A) and (B)?</li> <li>Are the options formatted as word...word?</li> </ul>                                                |
| The item that differs from the others (P22)                                                                                                  | <ul style="list-style-type: none"> <li>Are numbers (1) to (5) labeled in the text?</li> </ul>                                                                                                                                                                                                                                      |

Accuracy was 73.0% when *rank*,  $\alpha = \{16, 32\}$ ,  $\{32, 32\}$  without quantization, which represents a 58.7% improvement compared to the 46% of LLaMA-3-8B-Instruct without instruction fine-tuning, but 17.9% lower than the 88.9% of GPT-4o.

### C. OPPORTUNITIES AND RISKS

The proposed system presents several practical benefits. It allows for controllable question difficulty through prompt

engineering, enables rapid and cost-effective generation of a large number of questions compared to traditional item-writing methods, and serves as a valuable auxiliary tool for educators by producing practice materials.

However, the dataset generated by GPT-4 may contain inherent biases, which is a critical concern in high-stakes educational contexts. Additionally, the answers to the generated questions are not always guaranteed to be correct. These issues highlight the need for rigorous validation and



TABLE 7. Evaluation Criteria about answer generation. The numbers in parentheses indicate the number of task types.

| Task type (21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Evaluation Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose of the text (E1),<br>The author’s claim (E2),<br>Main point of the text (E3),<br>Subject of the text (E4),<br>Title of the text (E5),<br>Changes in emotion (E6),<br>Meaning of the underlined sentence (E7),<br>Inconsistencies in the table content (E8),<br>Inconsistencies in the content about the character (E9),<br>Consistencies in the content of the notice (E10),<br>Inconsistencies in the content of the notice (E11),<br>Grammar (E12),<br>Words that are contextually inappropriate (E13),<br>Choosing the correct word for the context (E14),<br>Fill in the blank (E15),<br>Words to fill in blanks A and B (E16),<br>Sentence that does not fit the flow (E17),<br>Order of the sentences (E18),<br>The most appropriate place for the sentence (E19),<br>Words to fill in summary blanks A and B (E20),<br>The item that differs from the others (E21) | <ul style="list-style-type: none"><li>• Are the translation and commentary formats printed separately?</li><li>• Did the translation include commentary?</li><li>• Did the commentary include any translations?</li><li>• Do the question type and answer type match?</li><li>• Does the commentary indicate that there is only one prediction?</li><li>• Is the prediction provided in the commentary included in the actual question options?</li><li>• If the prediction number presented in the commentary matches the ground truth number, do the commentary corresponding to that number and the content of the options corresponding to the ground truth match?</li></ul> |

human oversight. Therefore, while the system shows strong potential, we believe it is currently more appropriate for generating supplementary practice problems rather than for direct use in official test construction. Such cautious and limited use can help ensure the educational reliability and fairness of the resulting content.

Assessing our system’s effectiveness in real educational settings is critical. As part of this ongoing effort, we have launched a publicly accessible web-based service (<http://csatmaker.site>), which allows users to interact with the system. We are currently collecting user feedback through this platform for future improvements. A more systematic, user-centered evaluation should be conducted to validate the system’s utility in authentic educational contexts.

VII. CONCLUSION

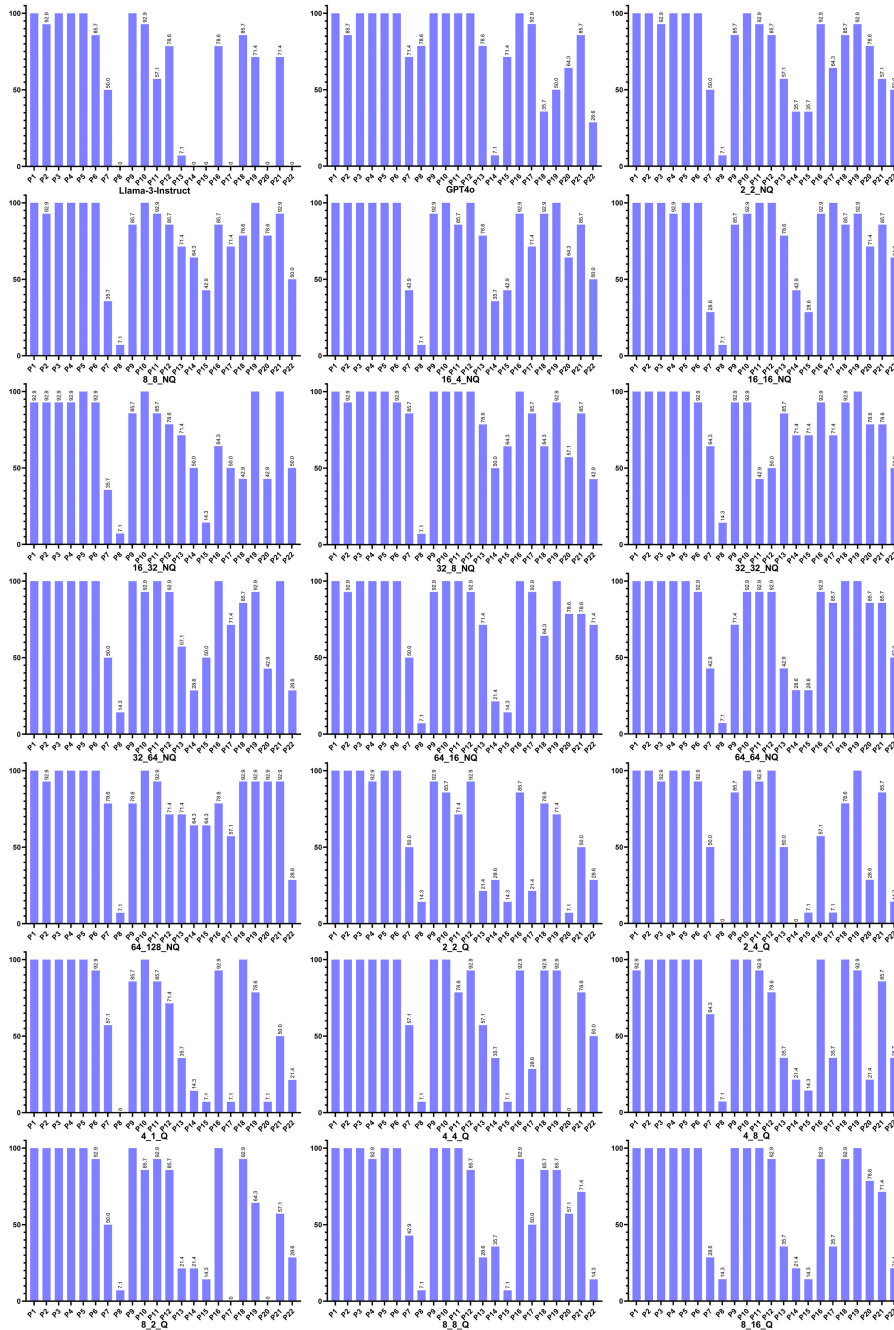
We introduced the KCSAT-ENG dataset, which contains KCSAT-ENG related questions and answers. To the best of our knowledge, no public dataset specific to KCSAT-ENG has been available. The dataset was carefully restructured and used as input for our LLMs, and extensive experiments were conducted by varying the rank, alpha, and quantization of QLoRA while performing instruction fine-tuning. We proposed new evaluation criteria based on the task types of KCSAT-ENG questions and answers. We implemented a KCSAT-ENG question and answer generation model using LLaMA-3-8B-Instruct as the backbone model, incorporating QLoRA and FlashAttention-2. Our key innovation lies in the use of two separate models for question and answer generation, enabling accuracy improvements through a cross-verification process. The experimental results demonstrated that training with the instruction fine-tuning method proposed in this paper significantly improved performance

compared to a model without instruction fine-tuning. Our findings revealed that the performance metrics did not exhibit a linear relationship with changes in rank and alpha.

Despite these promising outcomes, our dataset may still contain inherent biases introduced by GPT-4-generated content, highlighting the need for rigorous data validation and human oversight in high-stakes educational contexts. While this study focuses on KCSAT-ENG, the proposed approach can be adapted to other standardized exams or real-time learning platforms through careful reconfiguration of data collection, prompt engineering, and fine-tuning protocols. Future work may incorporate more diverse corpora or retrieval-augmented strategies to mitigate domain shift and maintain the model’s relevance over time. By addressing these potential biases and limitations, parameter-efficient techniques such as QLoRA can be extended to a broader range of domains and real-time applications, ultimately improving their robustness and practical utility. We hope this research serves as a valuable reference for future studies on KCSAT-style question and answer generation and beyond. Furthermore, the recently introduced DeepSeek R1 Distill appears to be a promising candidate for comparison due to its integration of reasoning strategies such as Chain-of-Thought during the RLHF stage. However, directly comparable evaluation metrics for LLaMA-3-8B-Instruct are not available, and DeepSeek is primarily optimized for English and Chinese. A rigorous comparison on Korean-language reasoning is left as future work.

APPENDIX A

Table 5 presents a complete example of the actual definitions used in this study.

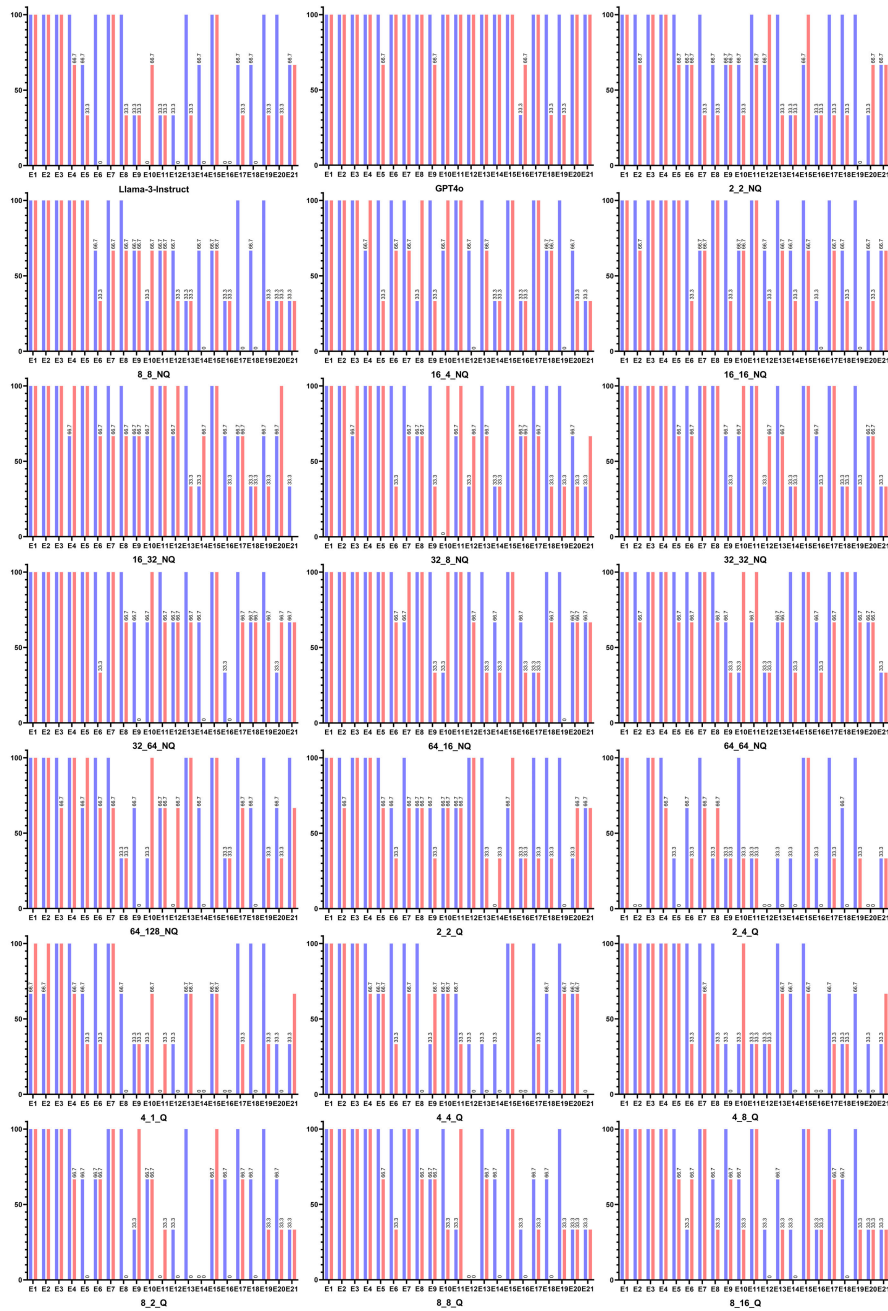


**FIGURE 9.** This table shows the format verification results for each type of the question generation model. The blue bars represent the format verification results, while the red bars indicate the accuracy. For items with a ratio of 100%, the values above the bars are omitted.

## APPENDIX B

The evaluation focuses on whether the generated questions and answers accurately follow the definitions. This appendix experiment evaluates only whether the output format is correct, excluding the educational aspects of the generated questions and answers. The evaluation criteria for the LLaMA-3-8B-Instruct model fine-tuned with KCSAT-

ENG-Question Generation are provided in Table 6. The evaluation criteria for the LLaMA-3-8B-Instruct model fine-tuned with KCSAT-ENG-Answer Generation are provided in Table 7. If any of the evaluation criteria are not met, it is classified as a format error (false or 0). However, for the evaluation of the LLaMA-3-8B-Instruct model fine-tuned with KCSAT-ENG-Answer Generation, the “Inconsistencies



**FIGURE 10.** Format verification and accuracy rate for each question type in the answer generation model.

in the content about the character” type was excluded from the evaluation as the corresponding test data could not be obtained.

## APPENDIX C

Fig. 10 shows the format verification and accuracy rate evaluation results for each question type of the answer generation model. The blue bars represent the format verification results, while the red bars indicate the accuracy.

For items with a ratio of 100%, the values above the bars are omitted.

## REFERENCES

- [1] W. Xin Zhao et al., “A survey of large language models,” 2023, *arXiv:2303.18223*.
- [2] J. Wei, M. Bosma, V. Y. Zhao, K. Guu, A. W. Yu, B. Lester, N. Du, A. M. Dai, and Q. V. Le, “Finetuned language models are zero-shot learners,” 2021, *arXiv:2109.01652*.
- [3] S. Zhang, L. Dong, X. Li, S. Zhang, X. Sun, S. Wang, J. Li, R. Hu, T. Zhang, F. Wu, and G. Wang, “Instruction tuning for large language models: A survey,” 2023, *arXiv:2308.10792*.

- [4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," 2020, *arXiv:2005.11401*.
- [5] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," 2021, *arXiv:2106.09685*.
- [6] T. K. F. Chiu, Q. Xia, X. Zhou, C. S. Chai, and M. Cheng, "Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education," *Comput. Education: Artif. Intell.*, vol. 4, Dec. 2022, Art. no. 100118.
- [7] S. e. a. Maghsudi, "Personalized education in the AI era: What to expect next," *IEEE Signal Process. Mag.*, vol. 38, no. 3, pp. 37–50, May 2021.
- [8] P. Denny, J. Prather, B. A. Becker, J. Finnie-Ansley, A. Hellas, J. Leinonen, A. Luxton-Reilly, B. N. Reeves, E. A. Santos, and S. Sarsa, "Computing education in the era of generative AI," *Commun. ACM*, vol. 67, no. 2, pp. 56–67, Jan. 2024.
- [9] H. Li, T. Xu, C. Zhang, E. Chen, J. Liang, X. Fan, H. Li, J. Tang, and Q. Wen, "Bringing generative AI to adaptive learning in education," Jun. 2024, *arXiv:2402.14601 [cs]*. [Online]. Available: <http://arxiv.org/abs/2402.14601>
- [10] I. Zulkernan, "Adoption of generative AI and large language models in education: A short review," in *Proc. Int. Conf. Electron., Inf., Commun. (ICEIC)*, Jan. 2025, pp. 1–5.
- [11] E. Latif, G. Mai, M. Nyaaba, X. Wu, N. Liu, G. Lu, S. Li, T. Liu, and X. Zhai, "AGI: Artificial general intelligence for education," 2023, *arXiv:2304.12479*.
- [12] Q. Li, L. Fu, W. Zhang, X. Chen, J. Yu, W. Xia, W. Zhang, R. Tang, and Y. Yu, "Adapting large language models for education: Foundational capabilities, potentials, and challenges," Apr. 2024, *arXiv:2401.08664 [cs]*. [Online]. Available: <http://arxiv.org/abs/2401.08664>
- [13] E. Kasneci et al., "ChatGPT for good? On opportunities and challenges of large language models for education," *Learn. Individual Differences*, vol. 103, Mar. 2023, Art. no. 102274.
- [14] L. Yan, L. Sha, L. Zhao, Y. Li, R. Martínez-Maldonado, G. Chen, X. Li, Y. Jin, and D. Gašević, "Practical and ethical challenges of large language models in education: A systematic scoping review," *Brit. J. Educ. Technol.*, vol. 55, no. 1, pp. 90–112, Aug. 2023.
- [15] C. Xiao, S. X. Xu, K. Zhang, Y. Wang, and L. Xia, "Evaluating reading comprehension exercises generated by LLMs: A showcase of ChatGPT in education applications," in *Proc. 18th Workshop Innov. Use NLP Building Educ. Appl. (BEA)*, 2023, pp. 610–625. [Online]. Available: <https://aclanthology.org/2023.bea-1.52>
- [16] Z. Lu, X. Li, D. Cai, R. Yi, F. Liu, X. Zhang, N. D. Lane, and M. Xu, "Small language models: Survey, measurements, and insights," 2024, *arXiv:2409.15790*.
- [17] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," 2023, *arXiv:2305.14314*.
- [18] K.-C. E. A. Chien. (2023). *Using Parameter Efficient Fine-tuning on Legal Artificial Intelligence*. [Online]. Available: <https://ceur-ws.org/Vol-3637/paper20.pdf>
- [19] Y. Cao, Z. Yao, Z. Chen, and Z. Deng, "CatMemo at the FinLLM challenge task: Fine-tuning large language models using data fusion in financial applications," 2024, *arXiv:2407.01953*.
- [20] C. Wang, M. Li, J. He, Z. Wang, E. Darzi, Z. Chen, J. Ye, T. Li, Y. Su, J. Ke, K. Qu, S. Li, Y. Yu, P. Liò, T. Wang, Y. Guang Wang, and Y. Shen, "A survey for large language models in biomedicine," 2024, *arXiv:2409.00133*.
- [21] Q. Liu, X. Wu, X. Zhao, Y. Zhu, D. Xu, F. Tian, and Y. Zheng, "When MOE meets LLMs: Parameter efficient fine-tuning for multi-task medical applications," 2023, *arXiv:2310.18339*.
- [22] C. Ling et al., "Domain specialization as the key to make large language models disruptive: A comprehensive survey," 2023, *arXiv:2305.18703*.
- [23] S. Hadzhikoleva, T. Rachovski, I. Ivanov, E. Hadzhikolev, and G. Dimitrov, "Automated test creation using large language models: A practical application," *Appl. Sci.*, vol. 14, no. 19, p. 9125, Oct. 2024. [Online]. Available: <https://www.mdpi.com/2076-3417/14/19/9125>
- [24] T. E. A. Gerald, "Introducing CQuAE : A new French contextualised question-answering corpus for the education domain," in *Proc. 2024 Joint Int. Conf. Comput. Linguistics, Lang. Resour. Eval. (LREC-COLING)*, 2024, pp. 9234–9244. [Online]. Available: <https://aclanthology.org/2024.lrec-main.808>
- [25] N. Mulla and P. Gharpure, "Automatic question generation: A review of methodologies, datasets, evaluation metrics, and applications," *Prog. Artif. Intell.*, vol. 12, no. 1, pp. 1–32, Mar. 2023, doi: [10.1007/s13748-023-00295-9](https://doi.org/10.1007/s13748-023-00295-9).
- [26] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics*, 2002, pp. 311–318. [Online]. Available: <https://aclanthology.org/P02-1040>
- [27] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Text Summarization Branches Out*, Jul. 2004, pp. 74–81. [Online]. Available: <https://aclanthology.org/W04-1013>
- [28] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. STATMT*, J. Goldstein, A. Lavie, C.-Y. Lin, and C. Voss, Eds., Jun. 2005, pp. 65–72. [Online]. Available: <https://aclanthology.org/W05-0909>
- [29] H. Saadany and C. Orasan, "BLEU, METEOR, BERTScore: Evaluation of metrics performance in assessing critical translation errors in sentiment-oriented text," in *Proc. Transl. Interpreting Technol. Online Conf. TRITON*, Jan. 2021, pp. 48–56, doi: [10.26615/978-954-452-071-7\\_006](https://doi.org/10.26615/978-954-452-071-7_006).
- [30] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating text generation with BERT," 2019, *arXiv:1904.09675*.
- [31] G. Biancini, A. Ferrato, and C. Limongelli, "Multiple-choice question generation using large language models: Methodology and educator insights," in *Adjunct Proc. 32nd ACM Conf. User Model., Adaptation Personalization*, Jun. 2024, pp. 584–590, doi: [10.1145/3631700.3665233](https://doi.org/10.1145/3631700.3665233).
- [32] S. Maity and A. Deroy, "The future of learning in the age of generative AI: Automated question generation and assessment with large language models," 2024, *arXiv:2410.09576*.
- [33] OpenAI et al., "GPT-4o system card," 2024, *arXiv:2410.21276*.
- [34] T. Dettmers and L. Zettlemoyer, "The case for 4-bit precision: K-bit inference scaling laws," 2022, *arXiv:2212.09720*.
- [35] T. Dao, "FlashAttention-2: Faster attention with better parallelism and work partitioning," 2023, *arXiv:2307.08691*.
- [36] *Hello Gpt-4o*. Accessed: Oct. 28, 2024. [Online]. Available: <https://openai.com/index/hello-gpt-4o>
- [37] Y. Wang et al., "Super-NaturalInstructions: Generalization via declarative instructions on 1600+ NLP tasks," 2022, *arXiv:2204.07705*.
- [38] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, "Chain-of-Thought prompting elicits reasoning in large language models," 2022, *arXiv:2201.11903*.
- [39] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "LLaMA: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.
- [40] meta. *Meta-llama/Meta-Llama-3-8B-Instruct*. Hugging Face. Accessed: Oct. 28, 2024. [Online]. Available: <https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>
- [41] EBS Dept. Educ. Broadcast. Editorial. (2022). *EBS CSAT Completion English Section English (2022)(2023 Preparing for the CSAT)*. [Online]. Available: <https://www.ebsi.co.kr/ebs/pot/pot/retrieveCourseDetailNw.ebs?bookId=LB00000004499>
- [42] Korea Mil. Acad. (2024). *Previous Exam Questions*. Accessed: Aug. 23, 2024. [Online]. Available: <https://www.kma.ac.kr>
- [43] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
- [44] T. B. Brown et al., "Language models are few-shot learners," 2020, *arXiv:2005.14165*.



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